

Hermes 3 3B — 50 Defining Details

Source: Technical Report (arXiv:2408.11857), HuggingFace model card, and independent analysis.
Compiled August 2026 on Jetson benchmarks.

Identity & Origin

1. Nous Research's first fine-tune in the 3B parameter class — all prior Hermes models were 8B+.
2. Full-parameter fine-tune of Llama-3.2 3B (not Llama 3.1 — the 8B/70B/405B used 3.1, but the 3B uses the newer 3.2 base).
3. Released August 2024 alongside the larger Hermes 3 family.
4. Authors: Ryan Teknium, Jeffrey Quesnelle, Chen Guang — the same team across the whole Hermes 3 line.
5. Trained on H100s via LambdaLabs GPU Cloud.
6. Open weights under the Llama 3 license — fully downloadable, no gated access.

Alignment Philosophy

7. Neutrally-aligned — no built-in moral refusals; guardrails belong in the system prompt, not the weights.
8. No such thing as latent thoughtcrime — the model does not judge the intent behind a request.
9. Highly steerable — adapts to whatever worldview the system prompt establishes.
10. Amnesia Mode — with a blank system prompt and Who are you?, the 405B exhibits emergent unfiltered behavior; the 3B inherits this steerability architecture.
11. System prompt sensitivity is a feature, not a bug — the model deliberately places itself within the system prompt's frame.

Training Pipeline

12. Two-phase post-training: SFT then DPO (Direct Preference Optimization).
13. ~390 million token SFT dataset — primarily synthetically generated.
14. 69% output tokens / 31% input tokens — heavily response-weighted, distilling from frontier models.
15. Data curation: March to August 2024 — 5 months of iterative collection and filtering.
16. AdamW optimizer, weight decay 0.01, learning rate 7×10^{-6} (selected via hyperparameter sweep on 8B).
17. Modified Axolotl framework for training.
18. Weights & Biases for experiment tracking.
19. NEFTune (noisy embeddings) used during fine-tuning to improve instruction-following.
20. Flash Attention 2 with attention mask-free variable sequence packing.
21. 96% packing efficiency — only 4% padding tokens wasted.
22. Target sequence length: 8192 to match Llama 3.2's native context window.

Data Mixture Breakdown

- 23. General instructions: 60.6% — the largest single category.
- 24. Domain expert data: 12.8% — specialized knowledge domains.
- 25. Mathematics: 6.7%.
- 26. Roleplaying & creative writing: 6.1% — a Nous Research hallmark.
- 27. Coding & software development: 4.5%.
- 28. Tool use, agentic reasoning & RAG: 4.3% — small percentage, huge practical impact.
- 29. Steering & alignment: 2.5% — dedicated data for system-prompt following.

Filtering & Quality

- 30. Token length thresholds to balance conversation lengths.
- 31. Removal of refusals — no I can't help with that examples in training data.
- 32. Removal of improperly formatted responses.
- 33. Elimination of conversations with missing or empty turns.
- 34. Prioritization of outputs from the strongest frontier models (GPT-4 and others).

Agentic Capabilities

- 35. XML tags for structured output — trained to emit XML blocks for machine-parseable responses.
- 36. Scratchpads — intermediate processing steps visible to the user/developer.
- 37. Internal monologue generation — transparent decision-making chains.
- 38. Mermaid diagrams — can generate visual flowchart/diagram markup for complex explanations.
- 39. Step-labeled reasoning and planning — explicit numbered reasoning steps.
- 40. Function calling via JSON schema in system prompt — structured tool_call blocks with name and arguments.
- 41. RAG with citation tags — trained to cite retrieval sources inline.
- 42. Multi-step tool chaining — can plan a sequence of tool calls to accomplish a complex goal.
- 43. Hermes-specific reserved control tokens from the Llama 3.1/3.2 tokenizer — extra slots used for reasoning markers.

Prompt Format & Interface

- 44. ChatML prompt format (not Llama's native format) — <|im_start|>/<|im_end|> tokens for structured multi-turn dialogue.

45. JSON mode — generates schema-compliant JSON on demand when a schema is provided in the system prompt.

46. vLLM native support — `--tool-call-parser hermes` flag parses XML tool calls into OpenAI-compatible format.

Performance & Position

47. 128K token context window inherited from Llama 3.2 3B base.

48. Competitive with Llama 3.1 Instruct on benchmarks (ARC, BoolQ, HellaSwag, IFEval, Winogrande) with varying strengths.

49. Over 280 adapter models and 50 merges built on top of the 8B variant alone — the 3B extends this ecosystem.

50. Jetson benchmark: 20.6 tok/s generation at Q4_K_M on NVIDIA Jetson 8GB — ties with Llama 3.2 3B, confirming shared architecture with Nous's training layer on top.